

PROJECT AND TEAM INFORMATION

Project Title

NightLight:Auto- Illuminating Lamp with Light Sensing

Student / Team Information

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1. OBJECTIVE =>

The objective of this project is to design and develop an **automatic night lamp system** that:

- Detects ambient light intensity using a sensor
- Automatically switches ON the lamp in low-light conditions (night)
- Automatically switches OFF the lamp in bright conditions (day)

This system aims to:

- Reduce energy consumption
- Eliminate manual switching
- Demonstrate the concept of sensor-based automation

1. INTRODUCTION =>

With the advancement of electronics and embedded systems, **automation has become essential** in everyday life. Lighting systems are one of the major areas where automation can significantly improve efficiency.

A **Night Light Auto-Illuminating Lamp** is a simple electronic system that uses a **Light Dependent Resistor (LDR)** to sense environmental light. The system operates without human intervention and is widely used in:

- Smart homes
- Street lighting
- Industrial lighting systems
- Security systems

This project demonstrates the use of **analog sensing + transistor switching** to create an intelligent lighting solution.

2. THEORY BACKGROUND =>

3.1 Light Dependent Resistor (LDR)

An LDR is a passive component whose resistance changes with light intensity.

- Bright light → Resistance decreases ($\sim 1\text{k}\Omega$ – $10\text{k}\Omega$)
- Darkness → Resistance increases ($\sim 100\text{k}\Omega$ – $1\text{M}\Omega$)

This property is based on **photoconductivity**, where incident photons increase the number of charge carriers in the material.

3.2 Transistor as a Switch

A transistor (e.g., BC547) works as an electronic switch:

- **Cut-off Region** → Transistor OFF → No current flow
- **Saturation Region** → Transistor ON → Current flows

The transistor amplifies the small signal from the LDR circuit to drive the load (LED/relay).

3.3 Voltage Divider Principle

The LDR is used in a **voltage divider circuit**:

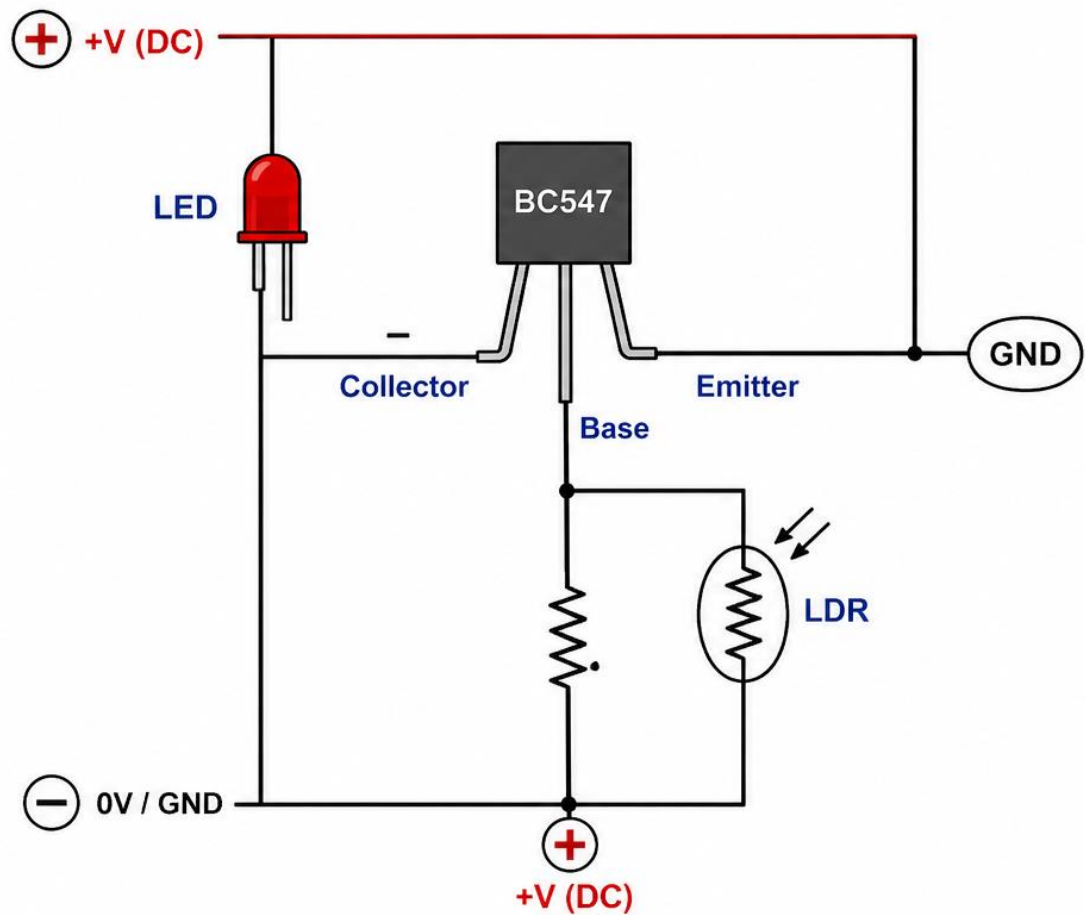
$$V_{out} = V_{cc} \times \frac{R}{R + R_{LDR}}$$

- Output voltage changes depending on light intensity
- This voltage controls the transistor

3. COMPONENTS REQUIRED =>

Component	Specification	Quantity
LDR	Light sensor	1
Transistor	BC547	1
Resistor	10kΩ, 1kΩ	1
LED / Bulb	Output load	1
Battery	9V	1
Breadboard	For connections	1
Connecting wires	—	As required

4. CIRCUIT DIAGRAM =>



5. WORKING PRINCIPLE =>

Case 1: Daytime (High Light Intensity)

- Light falls on LDR
- LDR resistance decreases significantly
- Voltage drop across LDR becomes small
- Base voltage of transistor is insufficient ($< 0.7V$)
- Transistor remains in **cut-off region**
- No current flows through collector

- LED/Bulb remains **OFF**
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Case 2: Nighttime (Low Light / Darkness)

- No light falls on LDR
- LDR resistance increases greatly
- Voltage across base increases
- Base-emitter junction becomes forward biased ($>0.7V$)
- Transistor enters **saturation region**
- Current flows from collector to emitter
- LED/Bulb turns **ON**

6. CONSTRUCTION STEPS =>

- >Place all components on breadboard
- >Connect LDR and resistor in series
- >Connect midpoint to transistor base
- >Connect emitter to ground
- >Connect collector to LED and resistor
- >Connect power supply
- >Test by covering and uncovering LDR

➔RESULT =>

The circuit operates successfully:

- Lamp turns ON automatically in darkness
- Lamp turns OFF automatically in light

The system responds accurately to changes in light intensity.

7. ADVANTAGES =>

- >Saves electrical energy
- >Fully automatic operation
- >Low cost and easy to build
- >Minimal maintenance
- >Can be expanded for large systems

9. DISADVANTAGES =>

- >LDR is temperature sensitive
- >Slow response compared to photodiodes
- >May malfunction under artificial light
- >Threshold not very precise

➔CONCLUSION:

The Night Light Auto-Illuminating Lamp is a simple yet effective application of electronics and automation. It demonstrates how sensor-based systems can be used to improve efficiency, reduce energy wastage, and provide convenience.

This project forms a strong foundation for understanding:

- Sensors
- Analog circuits
- Transistor switching
- Automation systems